

QuantMT5Live

Windows Build & MT5 Live-Trading Deployment Guide

Quant trading hackathon · MetaTrader 5 live execution (simulated funds, real matching)

Validated strategy recipe + competition-safe risk + resilient live loop, kill-switch & auto-reconnect

520 automated tests passing · order logic verified with a fake MT5 module

Read first — honest status. The order *logic* (lot sizing, BUY/SELL direction, position reconciliation, kill-switch, gates) is unit-tested with a fake MT5 module. The real `MetaTrader5.order_send` call has **not** been executed (that package is Windows-only). Validate on your Windows laptop using the §4 sequence *before* trading size. Run shadow mode first; it sends nothing.

1. Prerequisites (install once)

Component	Notes
Python 3.11 (64-bit)	python.org → tick "Add python.exe to PATH". Must match your MT5 bitness (64-bit).
MetaTrader 5 terminal	Install, log in to the competition account (Login / Password / Server from the organizer).
Git (optional)	Only if cloning from GitHub instead of copying the folder.

In MT5: **Tools** → **Options** → **Expert Advisors** → tick **"Allow algorithmic trading"**, and click the toolbar **"Algo Trading"** button so it is green. Keep the terminal open while the bot runs.

2. Get the code + create the environment

Put the `quantmt5live` folder on the laptop (copy it, or `git clone` your repo). Then open **PowerShell** in that folder:

```
py -3.11 -m venv .venv
.\.venv\Scripts\activate
python -m pip install -U pip
pip install -e ".[mt5]"      REM installs the package + MetaTrader5
```

3. Configure credentials

```
copy .env.example .env
notepad .env
```

Fill in `MT5_LOGIN`, `MT5_PASSWORD`, `MT5_SERVER`, and `MT5_TERMINAL_PATH` (e.g. `C:\Program Files\MetaTrader 5\terminal64.exe`). **Never commit** `.env` — it is git-ignored. If the broker uses suffixed symbols (e.g. `EURUSD.pro`), add `--mt5-symbol-map EURUSD=EURUSD.pro` (repeatable) to the run commands.

Verify the install (no orders placed)

```
quanthack check-environment
quanthack preflight
python -m unittest discover -s tests      REM expect: OK (520 tests)
```

4. Pre-go-live validation — DO THIS IN ORDER

1. **Read-only probe** — confirms login + live data, places nothing:

```
quanthack mt5-probe --confirm-read-only-mt5
```

2. **Shadow run** — computes & journals intended orders, **sends nothing**. Double-click `run_shadow.bat` (or run the command in §5 without `--i-understand-live-orders`). Watch `outputs\live_orders_journal.jsonl` : every line should be `MT5_SHADOW` with a sensible note ("would BUY 0.10 lots ..."). Sanity-check symbols, directions, lot sizes.
3. **One tiny manual order** — proves your broker accepts orders (independent path):

```
quanthack manual-ticket --symbol EURUSD --side BUY --target-notional 5000 --price <current-ask>
```

Place that small order once in MT5 and confirm it fills.

4. **Go live small** — set a tiny `--max-order-lots` (e.g. 0.05), then `run_live.bat` . Journal lines become `MT5_FILLED` ; confirm the position shows in MT5. Only then raise the cap.

5. The live command

```
quanthack live-trade --config configs\competition.toml --adapter mt5 ^
--poll-seconds 60 --iterations 100000 --max-order-lots 0.20 ^
--strategy-map EURUSD=macd_momentum --strategy-map XAUUSD=macd_momentum ^
--strategy-map GBPUSD=multi_horizon_momentum --strategy-map USDCHF=multi_horizon_momentum ^
--strategy-map USDCAD=macd_momentum --strategy-map AUDUSD=macd_momentum ^
--strategy-map EURGBP=volatility_squeeze --strategy-map EURCHF=volatility_squeeze ^
--strategy-map XAGUSD=macd_momentum --strategy-map USDJPY=quality_trend ^
--i-understand-live-orders
```

Without `--i-understand-live-orders` it runs in **shadow** mode. Disable Windows sleep so the laptop stays awake the whole round.

6. Safety features (built in)

Feature	What it does
Shadow default	No real order unless <code>--i-understand-live-orders</code> is passed.
RiskEngine gate	Margin floor, daily-loss stop, drawdown brake, leverage & per-symbol caps; blocked trades never reach the broker.
<code>--max-order-lots</code>	Hard cap on any single order (last-resort backstop).
Reconciliation	Repeated target sends only the delta — never stacks duplicate positions.
Resilient loop	A transient tick error is logged & skipped — the bot keeps running (avoids the >8h inactivity elimination).
Auto-reconnect	After a failed tick it closes + re-initializes the MT5 connection so a dropped link self-heals.
Kill-switch	<code>run_flatten.bat</code> / <code>quanthack mt5-flatten</code> closes ALL positions instantly.

Kill-switch (flatten everything)

```
quanthack mt5-flatten --config configs\competition.toml REM SHADOW (reports)
quanthack mt5-flatten --config configs\competition.toml --i-understand-live-orders REM LIVE (closes all)
```

7. Competition red-lines to respect

- **Log in within 8 hours of the 22:00 launch** or you are eliminated — start tonight.
- **Forced liquidation (30% margin level) = elimination.** The competition-safe config keeps leverage ~5x and margin level far above that; don't crank `--max-order-lots` toward the 28x discipline line.
- **API rate:** stay well under 500 req/sec. `--poll-seconds 60` is far below that.

8. Troubleshooting

Symptom	Fix
order fails with a filling-mode error	Switch <code>type_filling</code> to <code>ORDER_FILLING_FOK</code> or <code>RETURN</code> in <code>src/quanthack/trading/mt5_executor.py</code> . The journal <code>note</code> shows the broker <code>retcode/comment</code> .
"unknown instrument" / no ticks	Broker symbol is suffixed — add <code>--mt5-symbol-map EURUSD=EURUSD.pro</code> (and for metals/crosses too).
wrong lot sizes in shadow	Metals/crypto contract sizes vary by broker; verify shadow lots before going live.
bot stopped after reboot	Re-run <code>run_live.bat</code> . Consider Windows Task Scheduler to auto-restart.

Paper/simulated-funds competition tooling. Not financial advice. Validate the live path on Windows before trading size; the order round-trip was not executable on the build machine.